Summary in Graph

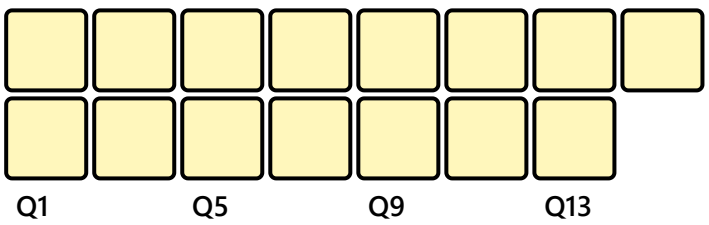
Exam Summary (GO Classes Test Series 2026 | Discrete Mathematics | Topic Wise Test 4)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (82) out of 818 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Numerical Type Award: 1 Penalty: 0 Combinatory

How many words can be formed by re-arranging the letters of the word "PROBLEMS" such that P and S occupy the first and last position respectively? (Note: The words thus formed need not be meaningful)

Your Answer: Correct Answer: 720 Not Attempted Time taken: 00min 00sec Discuss

Q #2 Numerical Type | Award: 1 | Penalty: 0 | Combinatory

Consider the set of 4-digit positive integers. How many of them have their digits in strictly increasing order? (i.e. $1234, 2578$ etc)

Your Answer: | Correct Answer: 126 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Numerical Type | Award: 1 | Penalty: 0 | Combinatory

How many ways are there to arrange the 12 letters of $\mathrm{AAABBBBCCCCC}$ without having two Cs together?

Your Answer: | Correct Answer: 1960 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Numerical Type | Award: 1 | Penalty: 0 | Combinatory

Consider the set $\mathrm{X} = \{2, 3, 4, 5, 6, 7, 8, 9\}$, which contains 8 elements. How many subsets of X have at least one prime number?

Your Answer: | Correct Answer: 240 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Numerical Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

Consider the set $\mathrm{X} = \{1, 2, 3, 4\}$ and $\mathrm{Y} = \{a, b, c, d, e, f\}$. How many one-to-one functions F are there from X to Y so that $\mathrm{F}(1)$ is not a vowel?

Your Answer: | Correct Answer: 240 | Not Attempted | Time taken: 00min 00sec | Discuss

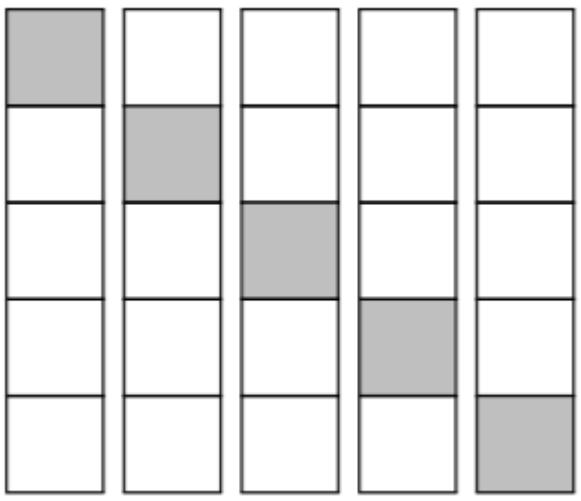
Q #6 Numerical Type | Award: 2 | Penalty: 0 | Combinatory

A student is given an exam consisting of 8 essay questions divided into 4 groups of 2 questions each. The student is required to select a set of 6 questions to answer, including at least 1 question from each of the 4 groups. How many sets of questions satisfy this requirement?

Your Answer: | Correct Answer: 24 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Numerical Type | Award: 2 | Penalty: 0 | Combinatory

Count the number of ways to place the numbers $1, 2, \ldots, 10$ (all of them) on the left and bottom side of the square below (below gray squares), such that for each gray box, the number on its left is smaller than the number below it.



Your Answer: | Correct Answer: 151200 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Numerical Type | Award: 2 | Penalty: 0 | Combinatory

Let the recurrence for the number of sequences of length n containing digits from $\{0, 1, \ldots, 6\}$, which do not have consecutive even digits be $\mathrm{T}(n)$. For instance, 05565301 is a sequence of length 8 containing digits from $\{0, 1, \ldots, 6\}$, which do not have consecutive even digits. We can see that $\mathrm{T}_1 = 7, \mathrm{T}_2 = 33$, which are the initial conditions of the recurrence $\mathrm{T}(n)$. For $n \geq 3, \mathrm{T}(n) = x\mathrm{T}_{n-1} + y\mathrm{T}_{n-2}$ Then what is $x + y$?

Your Answer: | Correct Answer: 15 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Numerical Type | Award: 2 | Penalty: 0 | Combinatory

Let $a_0, a_1, a_2, \ldots$ be a sequence with generating function $G(x) = \sum_{n=0}^{\infty} a_n x^n$, and let $b_0, b_1, b_2, \ldots$ be the sequence whose generating function is $F(x) = G(3x^2)$.

- Let $b_{11} = m.\, a_{11}$
- Let $b_{20} = p.\, a_{10}$

What is $m + p$?

Your Answer: | Correct Answer: 59049 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Numerical Type | Award: 2 | Penalty: 0 | Combinatory

How many integer solutions does the equation

$$x_1 + x_2 + x_3 + x_4 = 15$$

have, if we require that $x_1 \geq 2, x_2 \geq 3, x_3 \geq 10$ and $x_4 \geq -3$?

Your Answer: | Correct Answer: 20 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Select Type | Award: 2 | Penalty: 0 | Combinatory

Let $0 \leq \mathrm{S}(0) \leq \mathrm{T}(0)$, and suppose we have the recurrences

- $\mathrm{S}(n+1) = a\mathrm{S}(n) + f(n)$
- $\mathrm{T}(n+1) = b\mathrm{T}(n) + g(n)$,

where $0 \leq a \leq b$ and $0 \leq f(n) \leq g(n)$ for all $n \in \mathbf{N}$.

Which of the following is/are necessarily true ?

A. $\mathrm{S}(n) \leq \mathrm{T}(n)$ for all $n \in \mathbf{N}$.
B. $\mathrm{T}(n) \leq \mathrm{S}(n)$ for all $n \in \mathbf{N}$.
C. $\mathrm{T}(n) < \mathrm{S}(n)$ for some $n \in \mathbf{N}$.
D. $0 \leq \mathrm{S}(n) \leq \mathrm{T}(n)$ for all $n \in \mathbf{N}$.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 2 | Penalty: 0 | Combinatory

There are five people of different heights. In how many ways can they stand in a line, so there are no 3 consecutive people with increasing height(from left to right)? (Hint: May Try Inclusion-Exclusion Principle)

Your Answer: | Correct Answer: 70 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Combinatory

Let $n \in \mathbf{N}$ be a natural number and let $\mathrm{X} \subseteq \mathbf{N}$ be a subset with $n+1$ elements.

Then which of the following is/are necessarily true ?

A. there exist two distinct natural numbers $x, y \in \mathrm{X}$ such that $x-y$ is divisible by n.
B. there exist two distinct natural numbers $x, y \in \mathrm{X}$ such that both x, y are divisible by n.
C. there exist two distinct natural numbers $x, y \in \mathrm{X}$ such that $x+y$ is divisible by n.
D. there exist two distinct natural numbers $x, y \in \mathrm{X}$ such that xy is divisible by n.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 2 | Penalty: 0 | Combinatory

What is the coefficient of x^5 in the power series expansion of the following function?

$$\frac{1+7x}{(1-x)(1-5x)}$$

Your Answer: | Correct Answer: 9373 | Not Attempted | Time taken: 00min 01sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Combinatory

Let S be a set consisting of 10 elements. We want to count all the ways to form an unordered pair $\{\mathrm{A}, \mathrm{B}\}$ of disjoint subsets of S i.e. $\mathrm{A} \subseteq \mathrm{S}$, $\mathrm{B} \subseteq \mathrm{S}$, and A, B are disjoint sets. How many ways are there? Note that we will consider $\{\mathrm{X}, \mathrm{X}\}$ also as an unordered pair. Also, $\{\mathrm{X}, \mathrm{Y}\} = \{\mathrm{Y}, \mathrm{X}\}$.

Your Answer: | Correct Answer: 29525 | Not Attempted | Time taken: 00min 00sec | Discuss

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